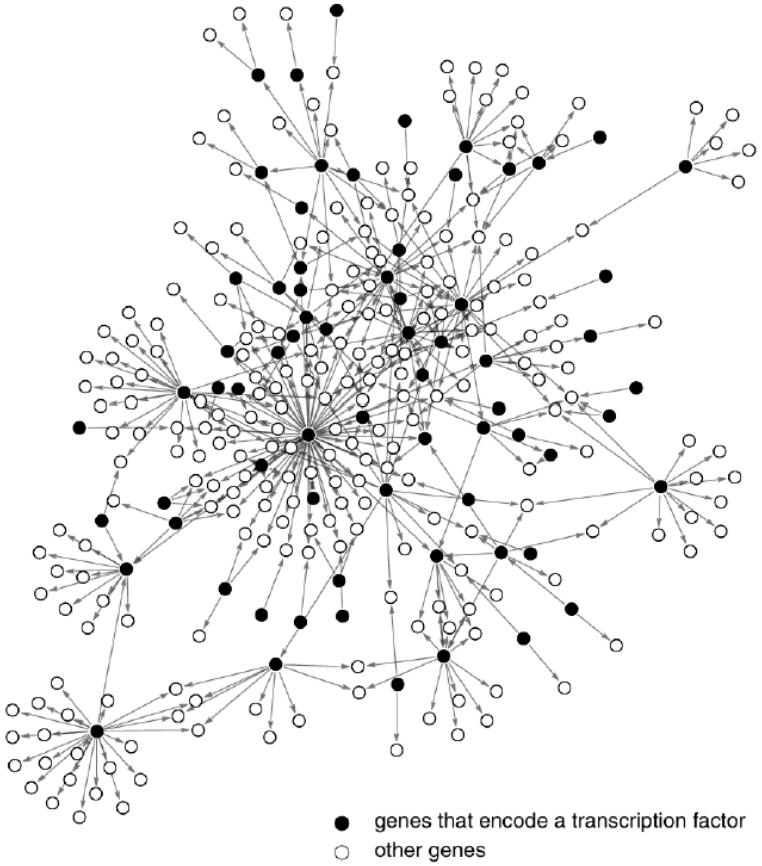
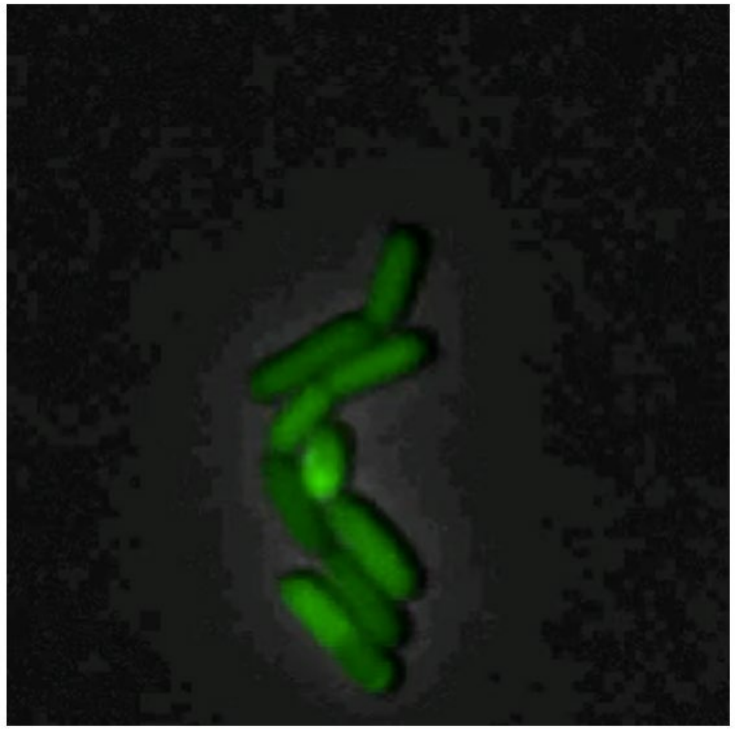
Gene regulatory networks (GRNs) provide a genome-wide view of cellular cognition



- N=4,500 nodes-genes
- E = 10,000 arrows

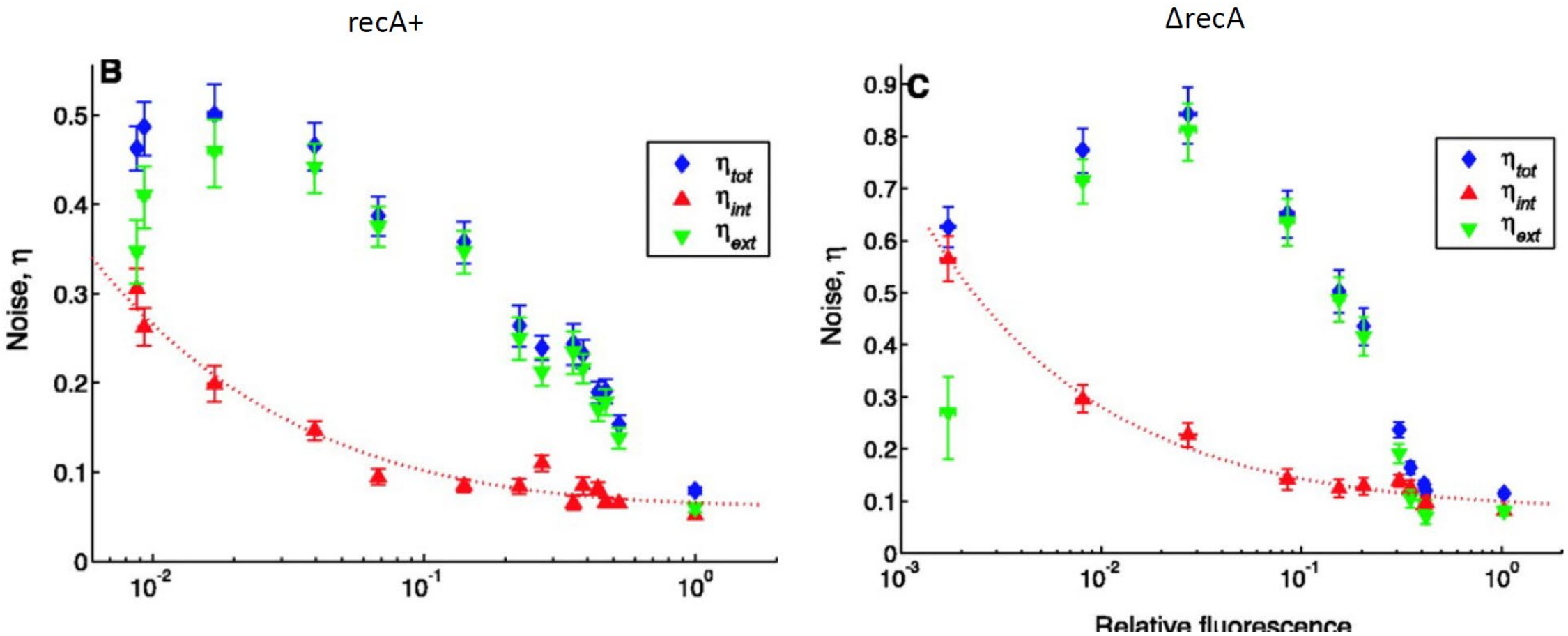
Uri Alon (2019) An Introduction to Systems Biology 2nd edition

But is gene expression noise meaningful biologically?



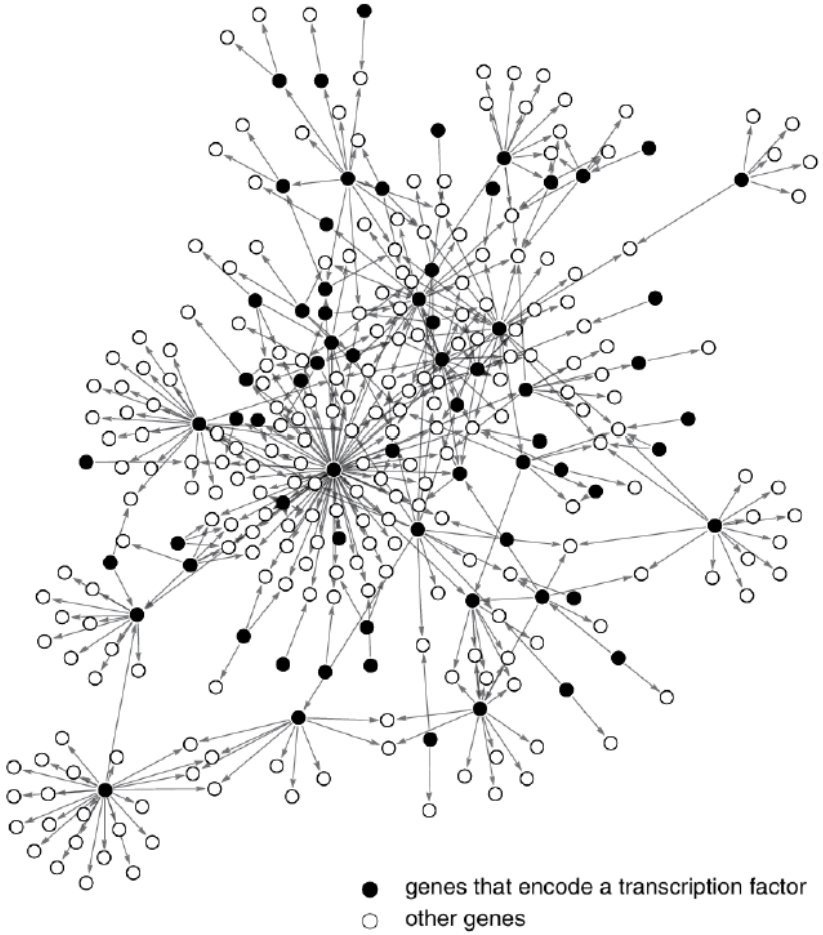
Locke JCW, Elowitz MB Nature Rev. Microbiol. (2009)

Elowitz et al measured intrinsic vs extrinsic noise in a beautiful experiment



Elowitz MB, Levine AJ, Siggia ED, Swain PS. Science (2002)

How do we model propagation of noise through chemical reaction networks that give rise to extrinsic noise?



Uri Alon (2019) An Introduction to Systems Biology 2nd edition

The chemical master equation (CME) gives an exact stochastic description of the reaction network

$$\varnothing \xrightarrow{R_+(n)} X, \qquad X \xrightarrow{R_-(n)} \varnothing.$$

$$n-1 \xleftarrow{R_-(n)} n \xrightarrow{R_+(n)} n+1$$

$$\frac{\partial P_n}{\partial t} = R_+(n-1)P_{n-1} + R_-(n+1)P_{n+1} - \left[R_+(n) + R_-(n)\right] P_n$$

Higher dimension chemical systems can be described by the rates vector and stoichiometry matrix

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = \sum_{j=1}^{R} S_{ij} r_j(\boldsymbol{x}) \qquad \Leftrightarrow \qquad \dot{\boldsymbol{x}} = S\boldsymbol{r}(\boldsymbol{x})$$

Elf & Ehrenberg, Genome Res. (2003)

But the CME is usually difficult to solve

$$\varnothing \xrightarrow{R_1^+(n_1)} X_1, \qquad X_1 \xrightarrow{R_1^-(n_1)} \varnothing$$

$$X_1 \xrightarrow{R_2^+(n_1,n_2)} X_1 + X_2, \qquad X_2 \xrightarrow{R_2^-(n_1,n_2)} \varnothing$$

$$\{n_1 - 1, n_2\} \xleftarrow{R_1^-(n_1)} \{n_1, n_2\} \xrightarrow{R_1^+(n_1)} \{n_1 + 1, n_2\},$$

$$\{n_1, n_2 - 1\} \xleftarrow{R_2^-(n_1,n_2)} \{n_1, n_2\} \xrightarrow{R_2^+(n_1,n_2)} \{n_1, n_2 + 1\}$$

$$\frac{\partial p_{\boldsymbol{n}}(t)}{\partial t} = \sum_{j=1}^{R} \left(\prod_{i=1}^{N} \mathcal{E}_i^{-S_{ij}} - 1 \right) \left(R_j p_{\boldsymbol{n}}(t) \right)$$

Paulsson, Nature (2004)

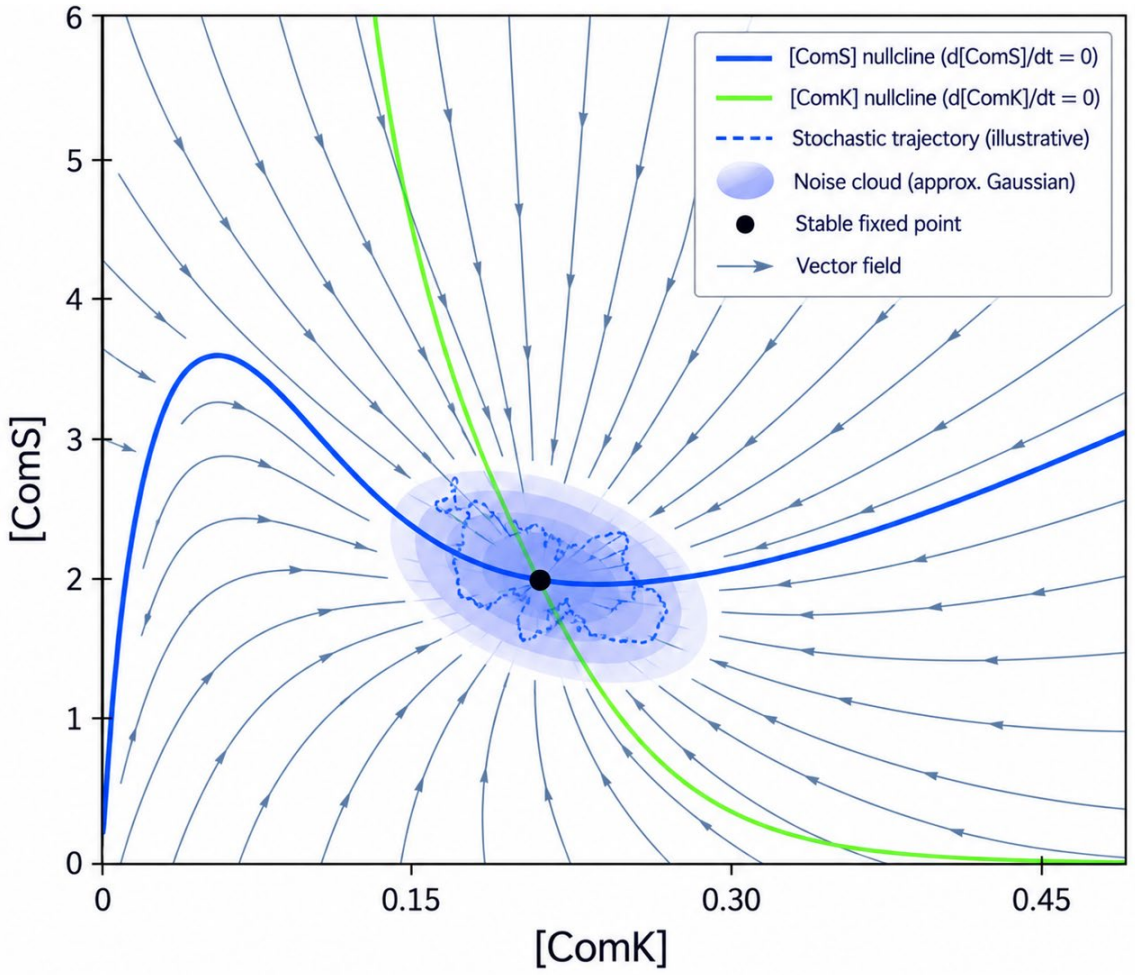
Van Kampen's Ω-expansion separates the CME into nonlinear deterministic dynamics and linear stochastic fluctuations

$$\frac{\partial p_{\boldsymbol{n}}(t)}{\partial t} = \sum_{j=1}^{R} \left(\prod_{i=1}^{N} \mathcal{E}_i^{-S_{ij}} - 1 \right) \left(R_j p_{\boldsymbol{n}}(t) \right)$$



Elf & Ehrenberg, Genome Res. (2003)

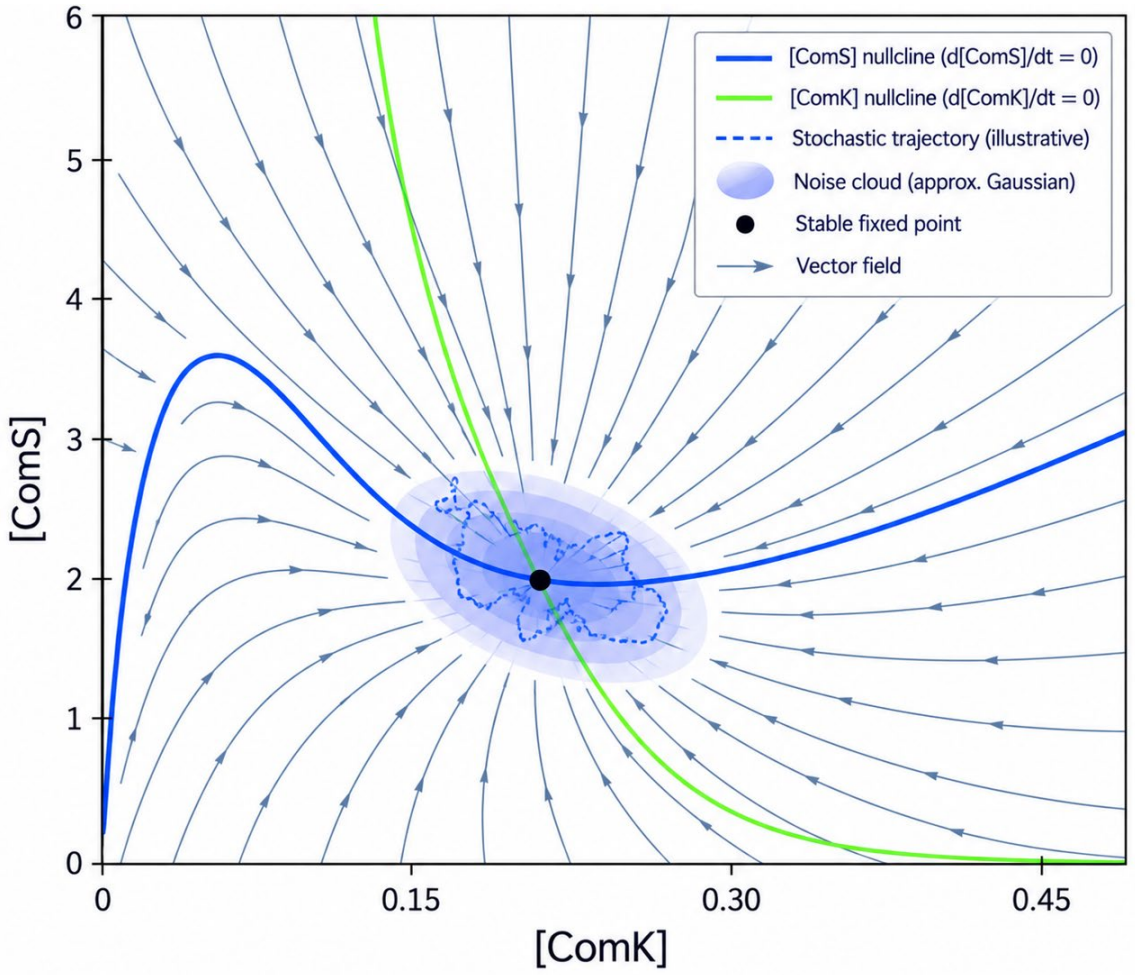
Separating nonlinear deterministic dynamics from Gaussian fluctuations lets us use familiar analytical tools



Süel GM, Garcia-Ojalvo J, Liberman LM & Elowitz MB, Nature 440, 545–550 (2006)

Deterministic dynamics restores and reshapes the cloud; stochastic reaction events continually broaden it;

Separating nonlinear deterministic dynamics from Gaussian fluctuations lets us use familiar analytical tools



Süel GM, Garcia-Ojalvo J, Liberman LM & Elowitz MB, Nature 440, 545–550 (2006)

At stationarity, the Lyapunov equation determines the covariance of the Gaussian fluctuations

$$\Pi_{\rm st}(\boldsymbol{\xi}) = \frac{\Omega^{N/2}}{(2\pi)^{N/2}\sqrt{\det\sigma}} \exp\left[-\frac{\Omega}{2}\boldsymbol{\xi}^T\sigma^{-1}\boldsymbol{\xi}\right]$$

$$A\sigma + \sigma A^T + \Omega B B^T = 0$$

$$A_{ik} = \sum_{j=1}^{R} S_{ij}\frac{\partial r_j}{\partial x_k} = \frac{\partial (S\boldsymbol{r})_i}{\partial x_k},$$

$$[BB^T]_{ik} = \sum_{j=1}^{R} S_{ij}S_{kj}r_j, \qquad BB^T = S\,\mathrm{diag}(\boldsymbol{r}(\boldsymbol{x}))S^T$$

Elf & Ehrenberg, Genome Res. (2003)

Rearranging the Lyapunov equation separates intrinsic and propagated noise in a simple gene network

$$\varnothing \xrightarrow{R_1^+(n_1)} X_1, \qquad X_1 \xrightarrow{R_1^-(n_1)} \varnothing$$

$$X_1 \xrightarrow{R_2^+(n_1,n_2)} X_1 + X_2, \qquad X_2 \xrightarrow{R_2^-(n_1,n_2)} \varnothing$$

$$H_{ij} = \left. \frac{\partial \ln(r_i^-/r_i^+)}{\partial \ln x_j} \right|_{x^*}$$

$$\frac{\sigma_2^2}{\langle n_2 \rangle^2} \simeq \underbrace{\frac{1}{\langle n_2 \rangle H_{22}}}_{\text{intrinsic noise}} + \underbrace{\frac{\sigma_1^2}{\langle n_1 \rangle^2}}_{n_1 \text{ noise}} \underbrace{\left(\frac{H_{21}}{H_{22}}\right)^2}_{\text{Susceptibility}} \underbrace{\frac{H_{22}/\tau_2}{H_{11}/\tau_1 + H_{22}/\tau_2}}_{\text{Time-averaging}}$$

Paulsson, Nature (2004)

Negative autoregulation suppresses noise by strengthening restoration and reducing susceptibility

$$H_{ij} = \left. \frac{\partial \ln(r_i^- / r_i^+)}{\partial \ln x_j} \right|_{x^*}$$

$$\frac{\sigma_2^2}{\langle n_2 \rangle^2} \simeq \underbrace{\frac{1}{\langle n_2 \rangle H_{22}}}_{\text{intrinsic noise}} + \underbrace{\frac{\sigma_1^2}{\langle n_1 \rangle^2}}_{n_1 \text{ noise}} \underbrace{\left(\frac{H_{21}}{H_{22}} \right)^2}_{\text{Susceptibility}} \underbrace{\frac{H_{22}/\tau_2}{H_{11}/\tau_1 + H_{22}/\tau_2}}_{\text{Time-averaging}}$$

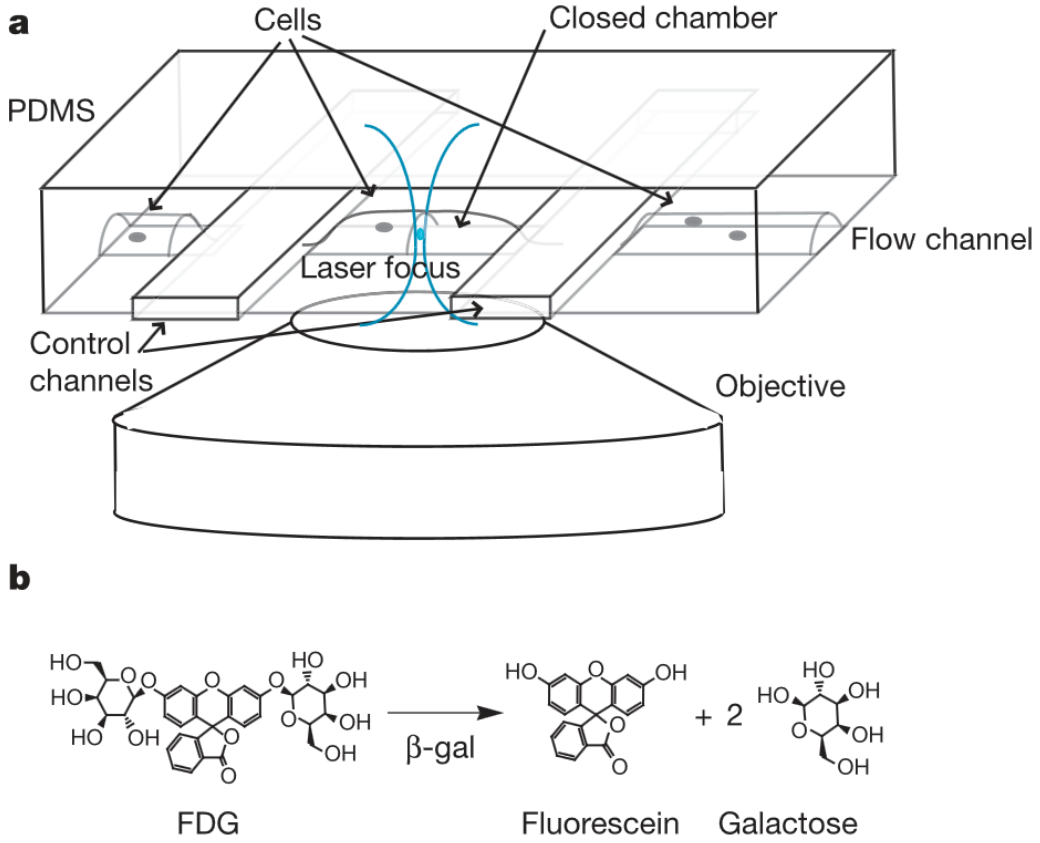
Paulsson, Nature (2004)

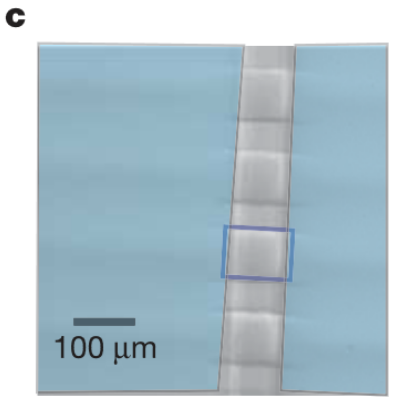
Short-lived mRNAs amplify transcriptional fluctuations into bursts of protein production

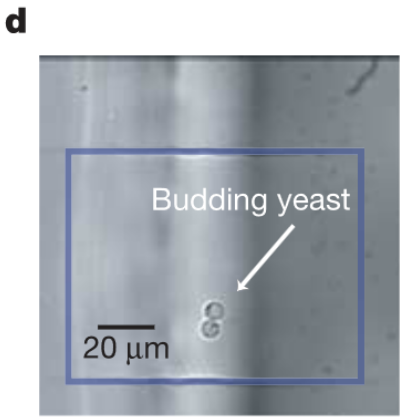
$$\frac{\sigma_2^2}{\langle n_2 \rangle^2} \simeq \underbrace{\frac{1}{\langle n_2 \rangle H_{22}}}_{\text{intrinsic noise}} + \underbrace{\frac{\sigma_1^2}{\langle n_1 \rangle^2}}_{n_1 \text{ noise}} \underbrace{\left(\frac{H_{21}}{H_{22}}\right)^2}_{\text{Susceptibility}} \underbrace{\frac{H_{22}/\tau_2}{H_{11}/\tau_1 + H_{22}/\tau_2}}_{\text{Time-averaging}}$$

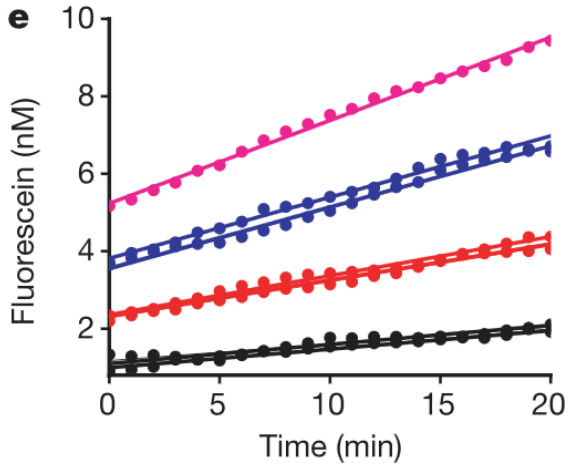
Paulsson, Nature (2004)

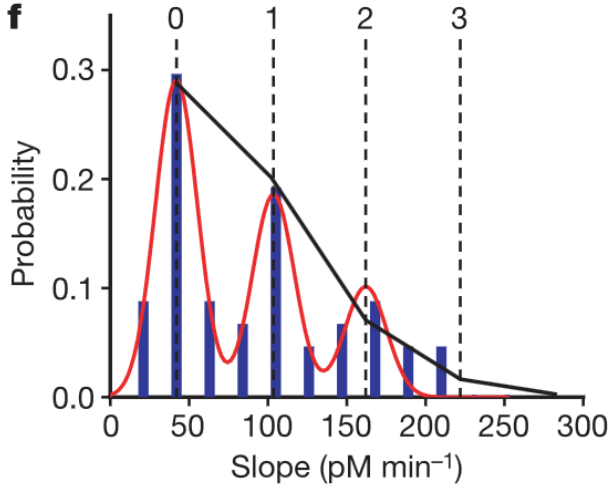
Cai, Friedman & Xie developed a system to count single proteins





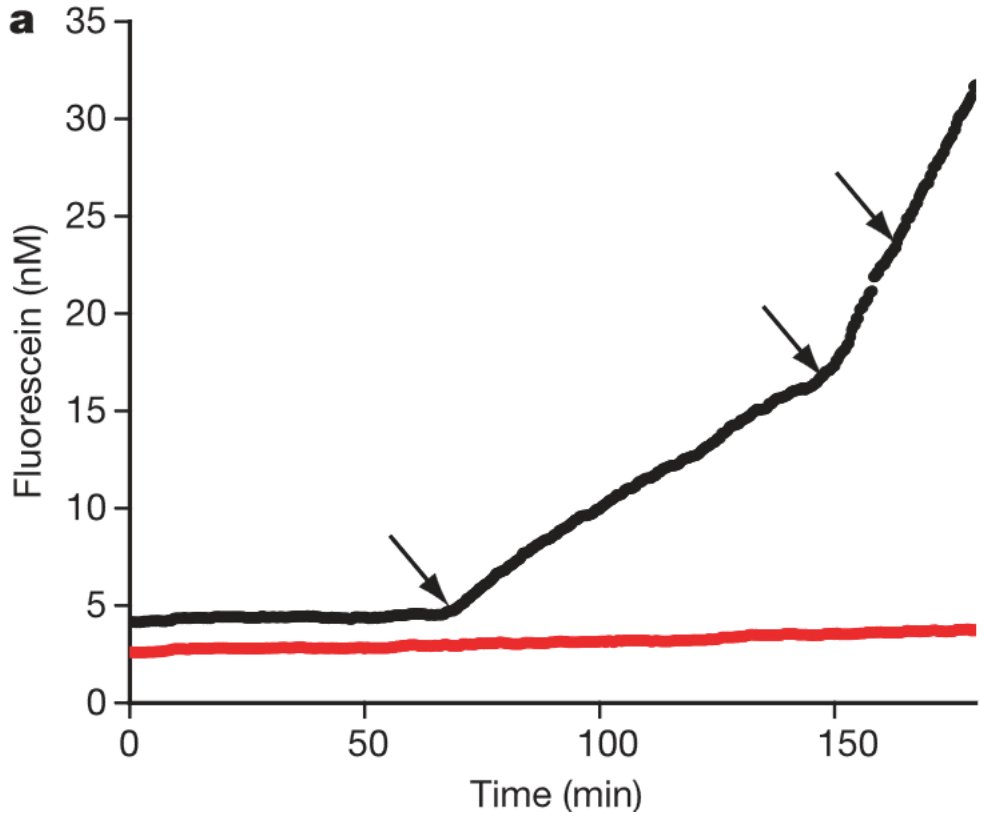


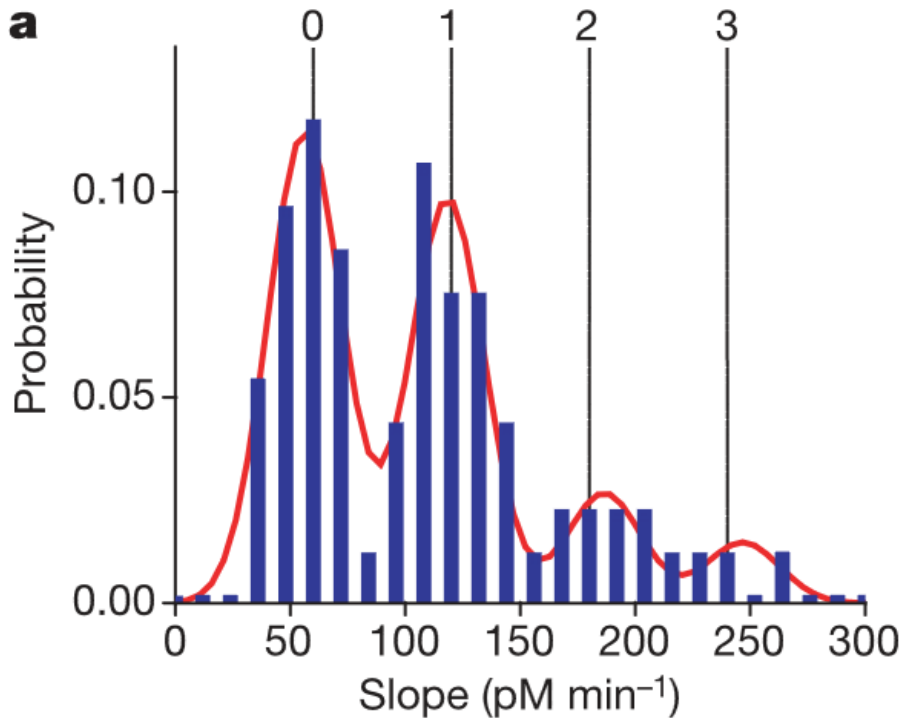




Cai L, Friedman N & Xie XS, Nature 440, 358–362 (2006)

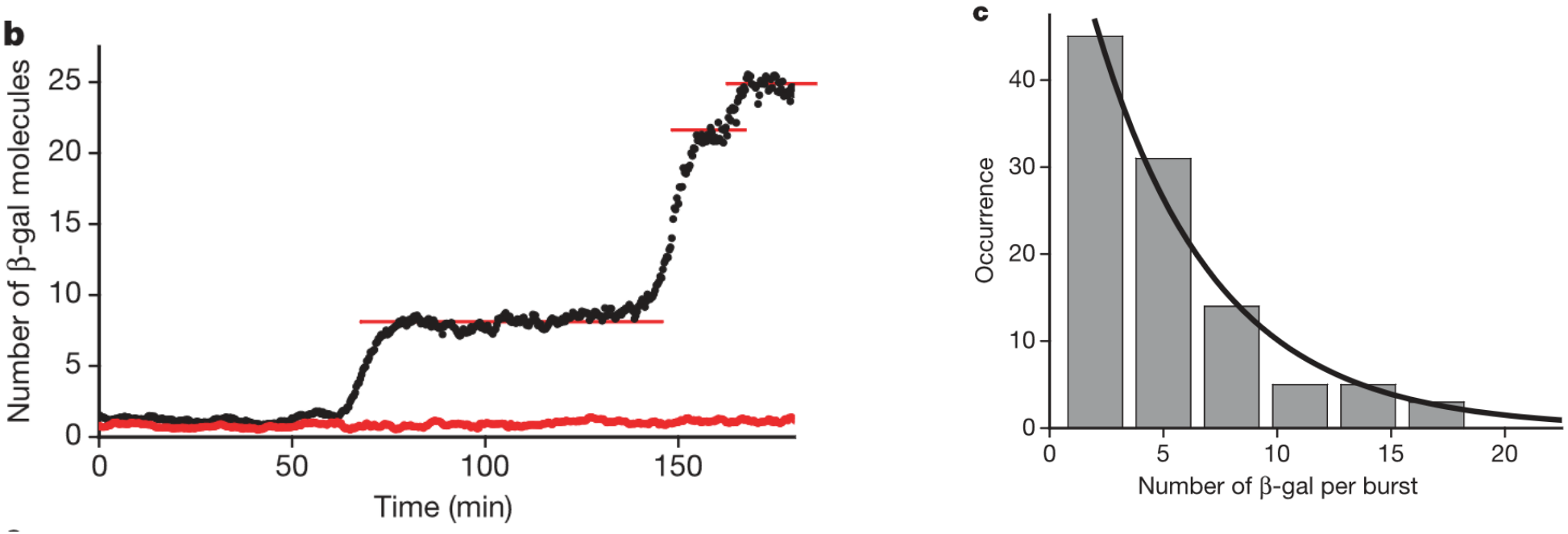
Individual protein molecules can be counted inside single cells





Cai L, Friedman N & Xie XS, Nature 440, 358–362 (2006)

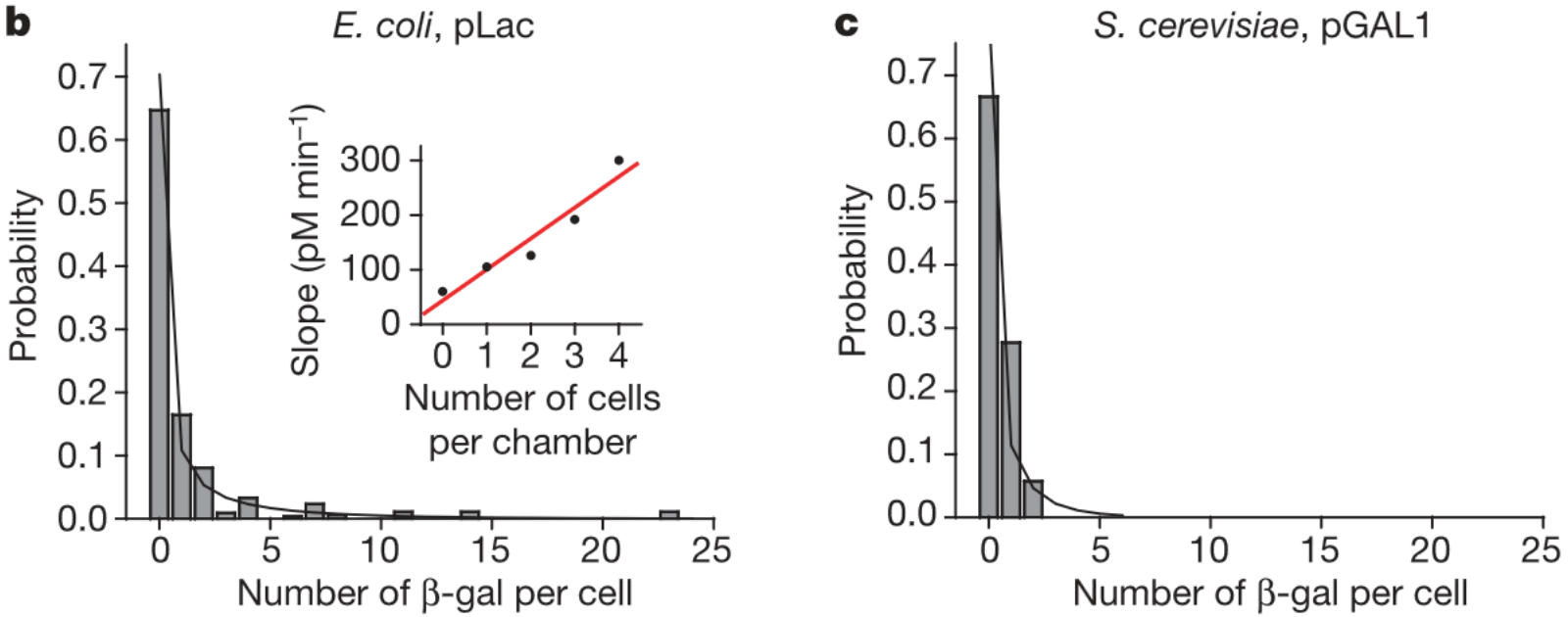
Proteins inside cells are produced in bursts

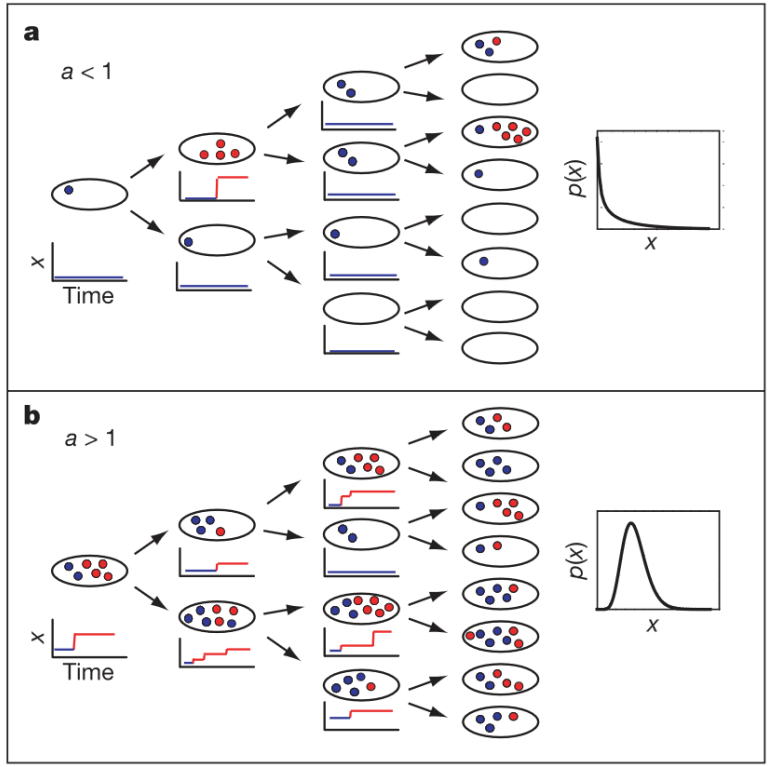


Cai L, Friedman N & Xie XS, Nature 440, 358–362 (2006)

Bursty protein production generates a Gamma distribution across cells

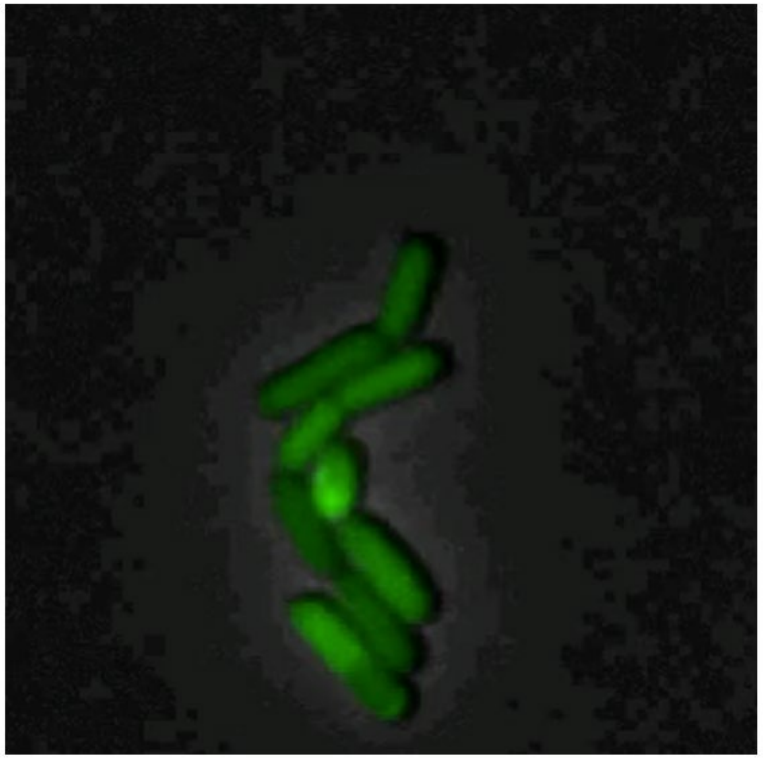
$$p(x) = \frac{x^{a-1} e^{-x/b}}{b^a \Gamma(a)}$$





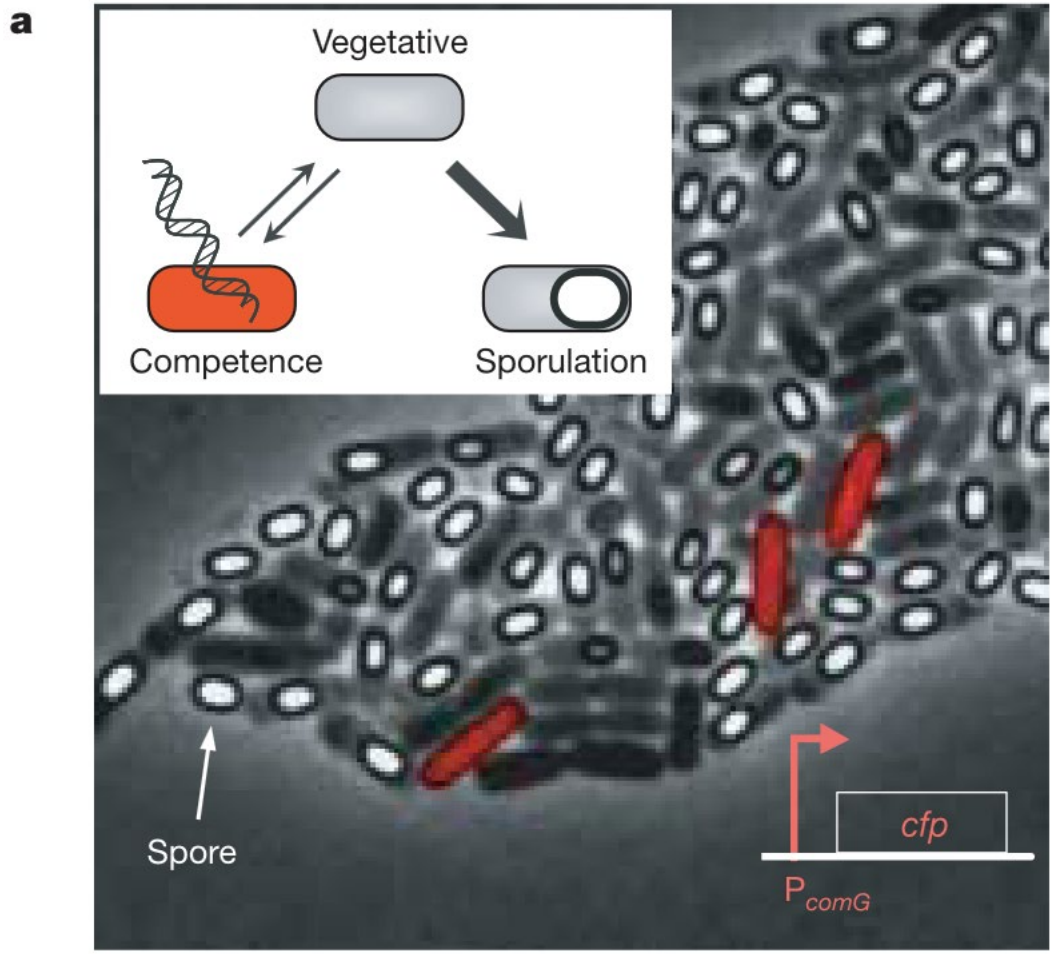
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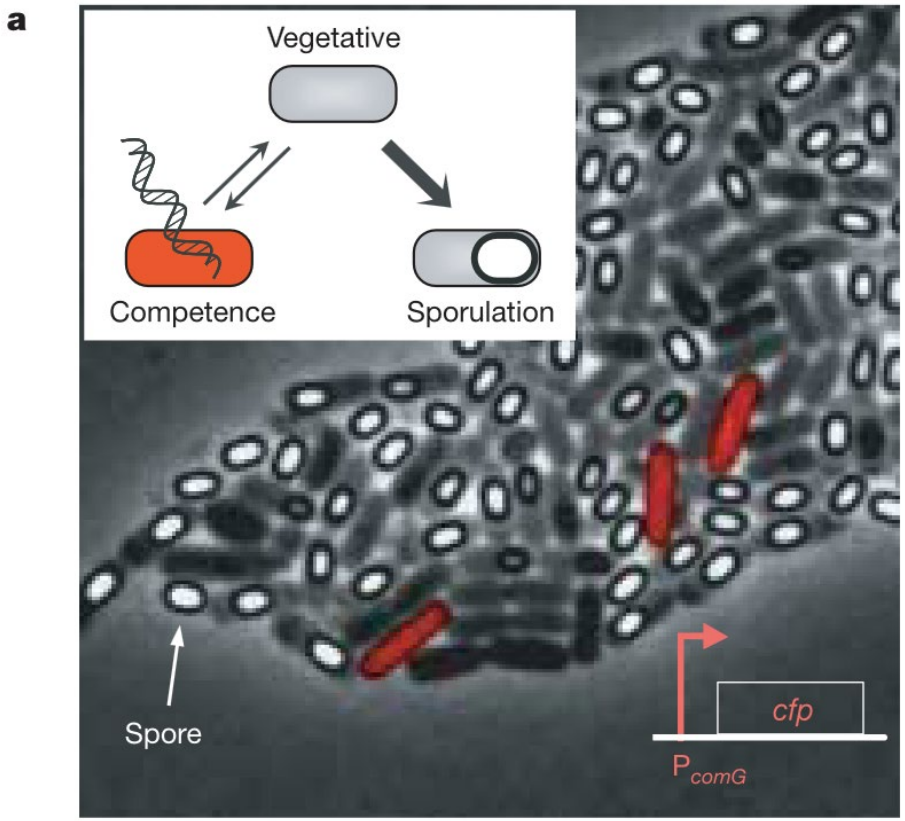
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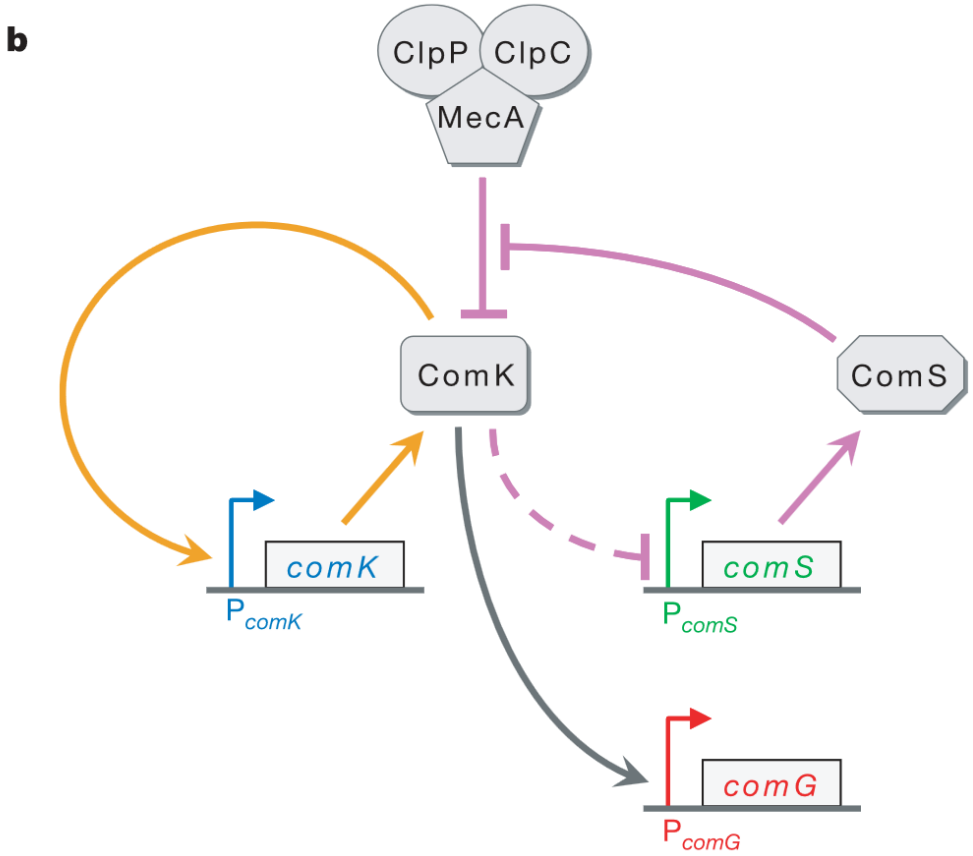
Molecular noise can have consequences at the scale of the whole cell



Süel GM, Garcia-Ojalvo J, Liberman LM & Elowitz MB, Nature 440, 545–550 (2006)

The master regulator ComK positively regulates itself, providing a mechanism to amplify molecular fluctuations





Süel GM, Garcia-Ojalvo J, Liberman LM & Elowitz MB, Nature 440, 545–550 (2006)